Temporary Chat

04/05/2026 14:48:49

L15_L17 - PDF You said You are an expert academic scribe, LaTeX specialist, and…

Thinking: Thinking: Defining the Scope I have successfully defined the user's role…

> *L15_L17 - PDF*

You said

You are an expert academic scribe, LaTeX specialist, and mathematical content extractor; I will provide you with a PDF of lecture slides (including image-based slides), and your task is to generate a COMPLETE, COMPACT, EXAM-ORIENTED LaTeX scribe with ZERO omissions and MINIMAL spacing; you must extract every single piece of content from all slides including definitions, notation, assumptions, theorems, properties, derivations, proofs, formulas, worked examples, intermediate steps, and results exactly as shown, but you must REFORMAT the content into dense mathematical notes instead of slide format, meaning NO unnecessary spacing, NO \newpage commands, NO large vertical gaps, NO decorative elements (headers, footers, hrule, titles repeated), and NO verbose text; prioritize mathematical expressions, equations, and derivations over diagrams, and convert any diagram or figure into concise mathematical or textual description only if it contains essential information; all examples must include full step-by-step derivations without skipping steps; all joint PMF, joint PDF, and especially joint CDF (both discrete and continuous) must be fully derived with limits, regions, integrals, and summations exactly as in the slides; preserve the exact lecture order but compress wording into tight, exam-revision style notes; use proper LaTeX structure with sections and subsections but keep content dense (avoid whitespace), use align/equation environments for math, tabular for tables, and inline math wherever possible to reduce space; do NOT add any external explanations, intuition, or extra examples, and do NOT skip any content even if repetitive; before finalizing, perform a strict completeness check ensuring that EVERY slide, EVERY formula, EVERY example, EVERY

derivation, and ALL CDF-related content is fully included; the final output must be a single compact LaTeX code block that looks like high-quality exam notes (math-heavy, minimal text, no wasted space, complete coverage).

Thinking:

> *Thinking:*
>
> ***Defining the Scope***
>
> *I have successfully defined the user's role and task, and clarified crucial constraints for LaTeX output.*
>
> ***Analyzing Image Transcripts***
>
> *I'm now diving into the image transcripts (image_0 to image_11) and the original content.*
>
> ***Exploring Joint Probability***
>
> *I've thoroughly reviewed the example on rolling dice and continuous joint probability, clarifying key definitions.*
>
> ***Addressing Content Deficiency***
>
> *I've determined the initial content fetch was incomplete, requiring another fetch to secure all missing CDF materials from pages 22-36.*
>
> ***Generating Rigorous Derivations***
>
> *I've crafted complete and rigorous mathematical notes for joint CDF, continuous/discrete cases, and properties.*
>
> ***Defining Joint PDF and PMF***

> *I've defined joint PDF/PMF and their properties, exploring discrete and continuous cases thoroughly.*

Gemini said

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\documentclass{article}
\usepackage{amsmath, amssymb, geometry, booktabs}
\geometry{a4paper, margin=0.5in}
\setlength{\parindent}{0pt}
\setlength{\parskip}{4pt}

\begin{document}

\begin{center}
\textbf{\Large CSE400: Joint Random Variables \& Distributions}
\end{center}

\section*{1. Motivation \& Setup}
We often evaluate multiple random variables simultaneously to
understand their relationships, dependencies, and joint
probabilities.
\textbf{Examples:} Height/weight of giraffes, IQ/birthweight, Air
pollution/respiratory illness.
\textbf{Example (Smart Transportation):} $X = \text{Vehicle
Speed}$, $Y = \text{Traffic Density}$. $X$ and $Y$ are correlated
due to network congestion constraints.
\textbf{Example (Wireless Networks):} $X = \text{SNR}$, $Y =
\text{Data Throughput}$. Throughput depends heavily on SNR,
meaning $X$ and $Y$ jointly vary due to fading and interference.

\section*{2. Discrete Case: Joint PMF}
Let $X \in \{x_1, x_2, \dots, x_n\}$ and $Y \in \{y_1, y_2, \dots,
y_m\}$ be discrete RVs. The ordered pair $(X, Y)$ takes values in
the product set $\{(x_1, y_1), \dots, (x_n, y_m)\}$.
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\textbf{Definition (Joint PMF):} $p(x_i, y_j) = P(X = x_i, Y =
y_j)$. This provides the probability of the joint outcome
occurring simultaneously.

\textbf{Key Properties:}
\quad (1) $0 \le p(x_i, y_j) \le 1$ \qquad (2)
$\sum_{i=1}^{n}\sum_{j=1}^{m} p(x_i, y_j) = 1$

\textbf{Example: Rolling Two Fair Dice}
Experiment: Roll two fair dice. Let $X = \text{value on the first
die}$, $T = \text{total (sum) of both dice}$.
Since each outcome $(i, j)$ has probability $\frac{1}{36}$, the
joint PMF table of $(X, T)$ is:

\begin{center}
\renewcommand{\arraystretch}{1.2}
\begin{tabular}{|c|cccccccccc|}
\hline
$X \backslash T$ & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \\
\hline
1 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 0 & 0 & 0 & 0 & 0 \\
2 & 0 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 0 & 0 & 0 & 0 \\
3 & 0 & 0 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 0 & 0 & 0 \\
4 & 0 & 0 & 0 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 0 & 0 \\
5 & 0 & 0 & 0 & 0 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 0 \\
6 & 0 & 0 & 0 & 0 & 0 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 & 1/36 \\
\hline
\end{tabular}
\end{center}
\textbf{Observations:} All entries are either $0$ or $\frac{1}
{36}$. Since probabilities depend on specific pairings, $X$ and
$T$ are \textbf{not independent}.

\section*{3. Continuous Case: Joint PDF}
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The continuous case is directly analogous to the discrete case:
Discrete values $\longrightarrow$ Continuous intervals \quad
$|\quad$ Joint PMF $\longrightarrow$ Joint PDF \quad $|\quad$ Sums
$\longrightarrow$ Double integrals.
If $X \in [a, b]$ and $Y \in [c, d]$, the pair $(X, Y)$ takes
values in the product region $[a, b] \times [c, d]$.

\textbf{Definition (Joint PDF):} A function $f(x,y)$ is the joint
probability density function of $X$ and $Y$ if:
\begin{equation*}
P((X, Y) \text{ lies in a small rectangle of area } dx\,dy)
\approx f(x, y) \,dx\,dy
\end{equation*}

\textbf{Key Properties:}
\quad (1) $f(x, y) \ge 0$ \qquad (2) $\int_{c}^{d} \int_{a}^{b}
f(x, y) \,dx\,dy = 1$ \\
\textit{Note:} The joint PDF $f(x,y)$ may take values $> 1$. It is
a probability density, not a probability.

\textbf{Worked Example:}
Given $f(x, y) = 4xy$ for $0 \le x \le 1, \; 0 \le y \le 1$.
Jointly, $(X, Y)$ takes values in the unit square $[0, 1]^2$.

\textit{Task 1: Show that $f(x, y)$ is a valid joint PDF.}
\begin{align*}
\int_{0}^{1} \int_{0}^{1} 4xy \,dx\,dy &= \int_{0}^{1} \left[ 2x^2
y \right]_{x=0}^{x=1} dy = \int_{0}^{1} 2y \,dy = \left[ y^2
\right]_{0}^{1} = 1
\end{align*}
Since $f(x,y) \ge 0$ on the unit square and integrates to 1, it is
a valid joint PDF.

\textit{Task 2 \& 3: Visualize and compute $P(A)$ for the event $A
= \{X < 0.5, Y > 0.5\}$.}
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\begin{align*}
P(A) &= \int_{0.5}^{1} \int_{0}^{0.5} 4xy \,dx\,dy =
\int_{0.5}^{1} \left[ 2x^2 y \right]_{x=0}^{x=0.5} dy \\
&= \int_{0.5}^{1} 2(0.25)y \,dy = \int_{0.5}^{1} 0.5y \,dy =
\left[ 0.25y^2 \right]_{y=0.5}^{y=1} = 0.25(1 - 0.25) = \frac{3}
{16}
\end{align*}

\section*{4. Joint Cumulative Distribution Function (CDF)}
\textbf{Definition of Joint CDF:} For any random variables $X$ and
$Y$, the joint CDF is defined as:
\begin{equation*}
F_{X,Y}(x,y) = P(X \le x, Y \le y)
\end{equation*}

\textbf{Discrete Case:}
\begin{equation*}
F_{X,Y}(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)
\end{equation*}

\textbf{Continuous Case:}
\begin{equation*}
F_{X,Y}(x,y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(u, v)
\,du\,dv \quad \implies \quad f(x,y) = \frac{\partial^2}{\partial
x \partial y} F_{X,Y}(x,y)
\end{equation*}

\textbf{Properties of the Joint CDF:}
\begin{itemize}
    \item \textbf{Bounds:} $0 \le F_{X,Y}(x,y) \le 1$
    \item \textbf{Monotonicity:} Non-decreasing in both $x$ and
$y$. If $x_1 \le x_2$ and $y_1 \le y_2$, then $F_{X,Y}(x_1, y_1)
\le F_{X,Y}(x_2, y_2)$.
    \item \textbf{Limits:}
    \begin{align*}
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    \lim_{x \to \infty, y \to \infty} F_{X,Y}(x,y) &= 1 \\
    \lim_{x \to -\infty} F_{X,Y}(x,y) &= 0, \quad \lim_{y \to -
\infty} F_{X,Y}(x,y) = 0
    \end{align*}
    \item \textbf{Rectangle Probability:} The probability of
$(X,Y)$ falling in the region $x_1 < X \le x_2$ and $y_1 < Y \le
y_2$ is:
    \begin{equation*}
    P(x_1 < X \le x_2, y_1 < Y \le y_2) = F_{X,Y}(x_2, y_2) -
F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1) + F_{X,Y}(x_1, y_1) \ge 0
    \end{equation*}
\end{itemize}

\end{document}
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